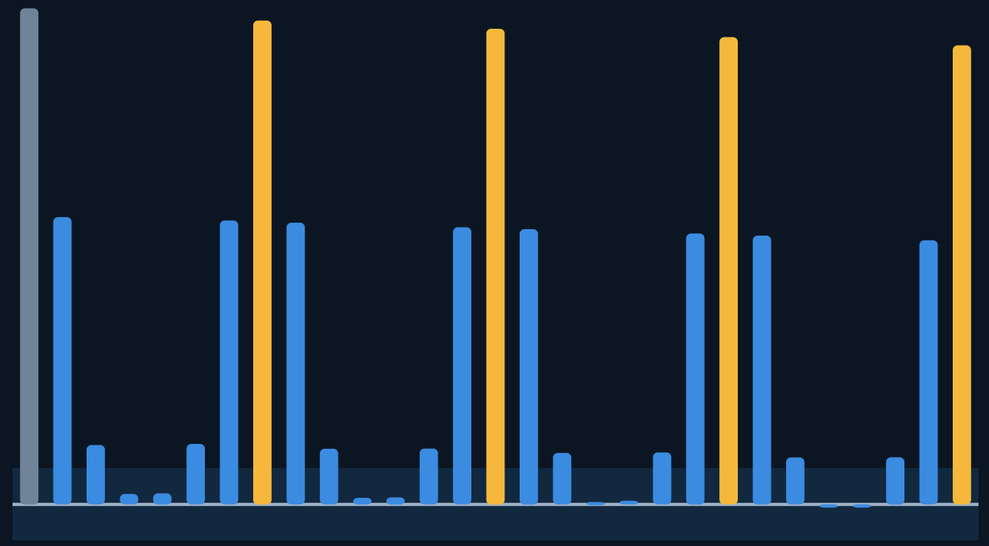
WEEK 2 · DAY 2

Does Today Remember Yesterday?

Autocorrelation: how to measure a series' memory, and how to spot a fake correlation. 🕵🏾



the “memory” of 2 years of satellite throughput

NIGCOMSAT
AFRICAN ROOTED GLOBALLY POSITIONED

Publica Academy

TODAY'S MISSION · 2 HOURS

Three moves



Shift a copy

line today up with yesterday (a "lag")



Measure the memory

autocorrelation, lag by lag



Read the correlogram

find the season; catch fake links

⏱️ 10 min hook · 🎤 30 min slides · 👩🏾‍💻 60 min hands-on · 🏁 20 min wrap-up

QUICK QUESTION

Third Mainland Bridge was jammed at 8am today. 🚗🚗🚗



Will it be jammed tomorrow at 8am?

Probably. Today tells you a lot about tomorrow. That's memory at **lag 1**.



Will it be jammed next Monday at 8am?

Even more likely! Mondays look like Mondays. That's memory at **lag 7**.

Today we measure that memory with one number. Your satellite data has it too. 🛰️

LAST TIME 👉 TODAY

Yesterday we were told the season. Today we find it ourselves.



Decomposition

You needed `period=7`, `12` or `24`. Somebody gave you the number.



Autocorrelation

The data **shows you** the period. No guessing, no copying from the code book.

PART 1

Memory 🧠

Line a series up with its own past and see how alike they are.

WHAT IS A LAG?

A copy of the series, pushed back in time

Satellite throughput (Gbps), the first 8 days of the file

day	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon
today s	3.19	3.14	3.30	3.25	3.14	2.47	2.33	3.18
yesterday `s.shift(1)`	NaN	3.19	3.14	3.30	3.25	3.14	2.47	2.33

```
s.shift(1)   # yesterday
s.shift(7)   # same weekday last week
```

Now every row has **today** and **yesterday** side by side.
Day 1 has no yesterday, so it gets **NaN**.

AUTOCORRELATION

How much does the series look like its own shifted copy?

One number between −1 and +1. “Auto” = with itself.




```
s.autocorr(lag=1)    # 0.58  today vs yesterday
s.autocorr(lag=7)    # 0.99  today vs same weekday last week
```

TODAY'S MAIN DATASET · BANDWIDTH_THROUGHPUT_DAILY.CSV

8 weeks of throughput: spot the rhythm



weekends

weekdays ≈ 3.3 Gbps

weekends ≈ 2.5 Gbps

synthetic teaching data · 730 days

THE MEMORY, IN NUMBERS

It remembers yesterday a little. It remembers last Monday a lot.

0.58	0.99	0.99	0.09
lag 1 yesterday	lag 7 last week	lag 14 2 weeks ago	lag 30 a month ago

Like your mama: she forgets what you said yesterday, but **never** forgets you skipped church last Sunday. 😄

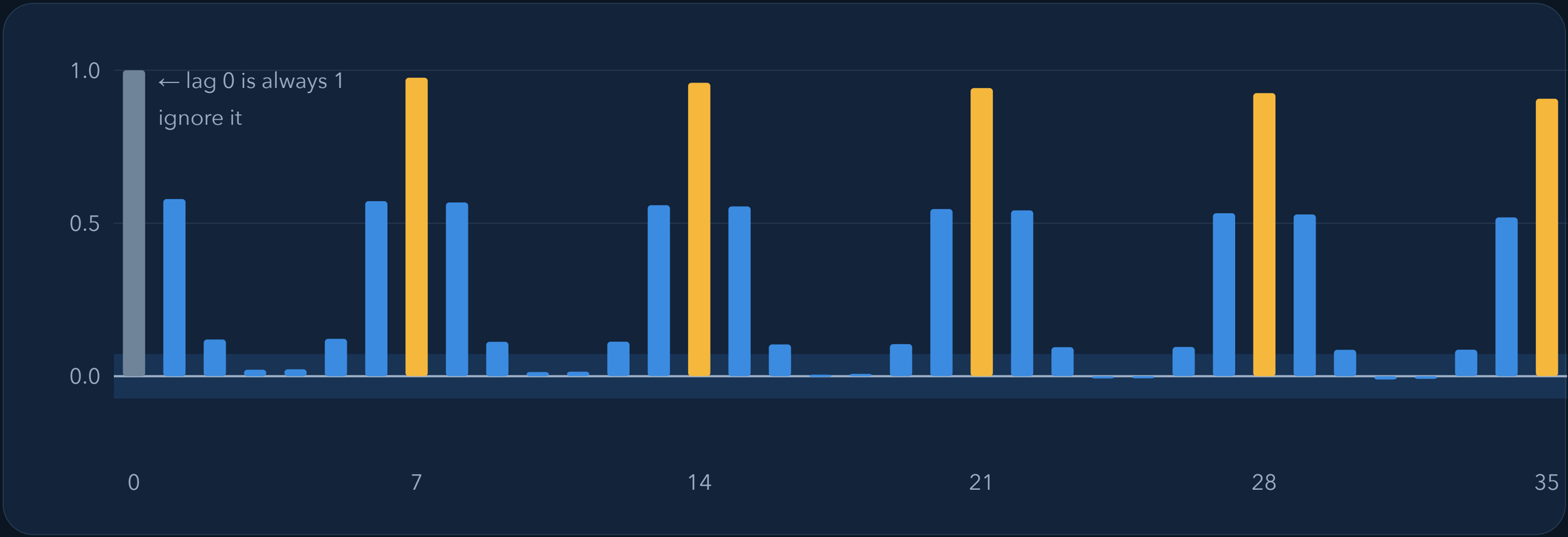
PART 2

The Correlogram 📊

Every lag at once, on one chart.

PLOT_ACF(S, LAGS=35)

The correlogram: one bar per lag



spikes at 7 · 14 · 21 · 28 · 35

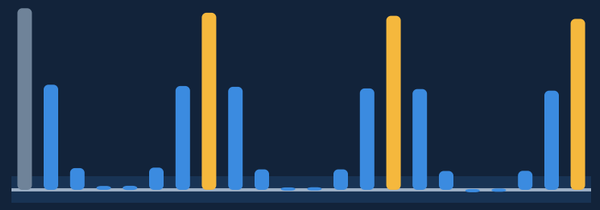
shaded band = could be chance

gap between spikes = the period

READING A CORRELOGRAM

Three shapes you'll meet

🎯 Regular spikes



= a **season**. Period = gap between spikes.

throughput, daily

🐢 Slow decay



= a **trend**. Every month looks like the last.

transponder use, monthly

🎲 All inside the band



= **no memory**. Pure noise.

illustration

FIND THE PERIOD FROM THE DATA

The spikes tell you the season length



Throughput

daily · spikes at 7, 14, 21

7

working week



NOC tickets

daily · lag 7 = 0.72

7

weekends are quiet



Battery charge

hourly · lag 24 = 1.00

24

eclipse at 21:00 every day



Ka-band

daily · peak at lag 365

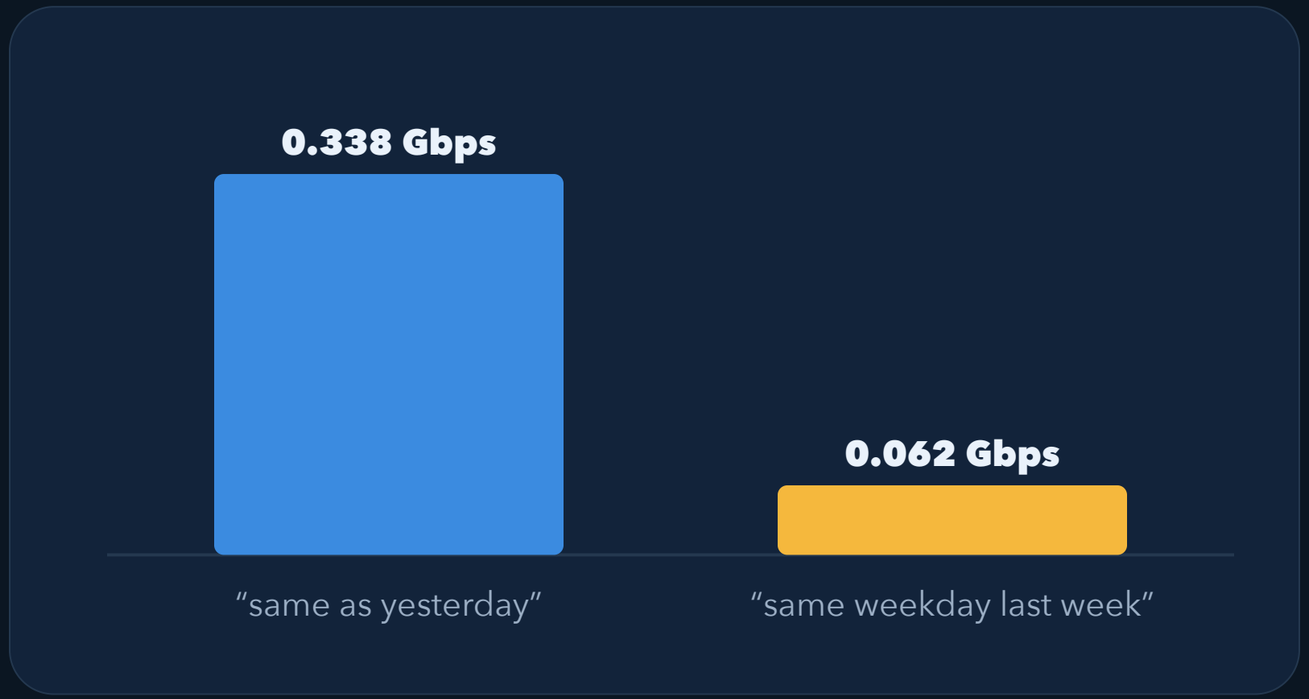
365

rainy season, yearly

The period is a property of **the process**, not a default setting. Check it, don't assume it.

SNEAK PEEK OF WEEK 5

Memory = a free forecast 🎁



Average daily error of two lazy forecasts on 2 years of throughput:

“Same weekday last week” is about **5× more accurate** than “same as yesterday”.

Because lag 7 (0.99) remembers more than lag 1 (0.58).

WHY MEMORY MATTERS FOR TESTING

Never split time data at random 🙅🏾

Random split: ■ train ■ test



Day 10 is in test, but days 3 and 17 (the same weekday, r = 0.99) are in train. **The answer leaked.**

Split by time: train on the past, test on the future



Like WAEC: if you saw the questions before the exam, your score is fake. 📝 ✅ Honest.

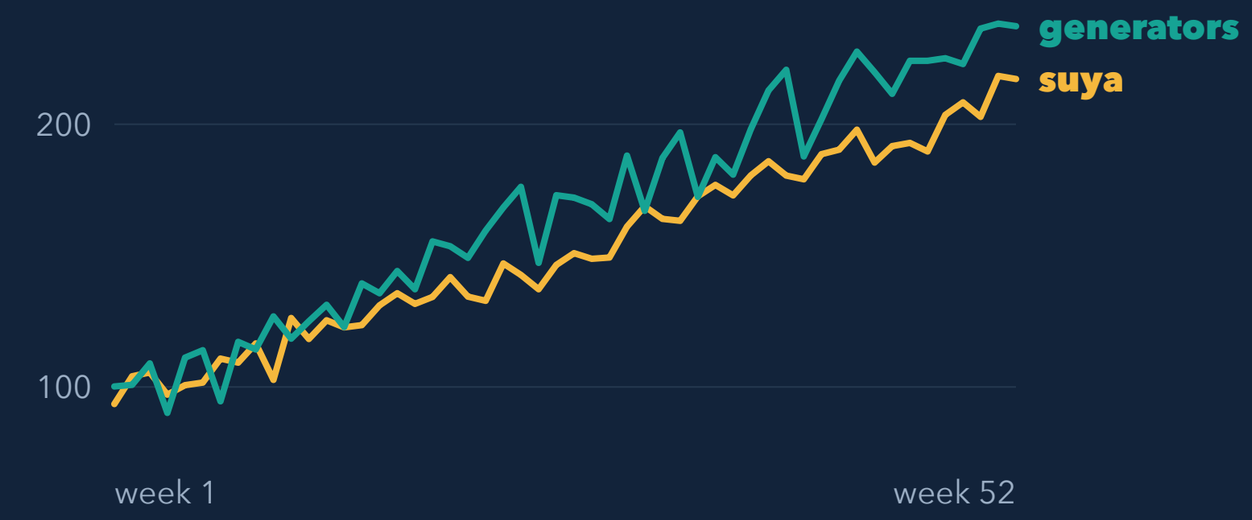
PART 3

Correlation is not cause 

When two series move together, ask why before you believe it.

A MADE-UP STORY FROM ONE LAGOS MARKET

Suya sales and generator sales rise together



weekly sales, indexed (first month = 100)

correlation of the levels

0.97

"Suya sells generators!" 🤔

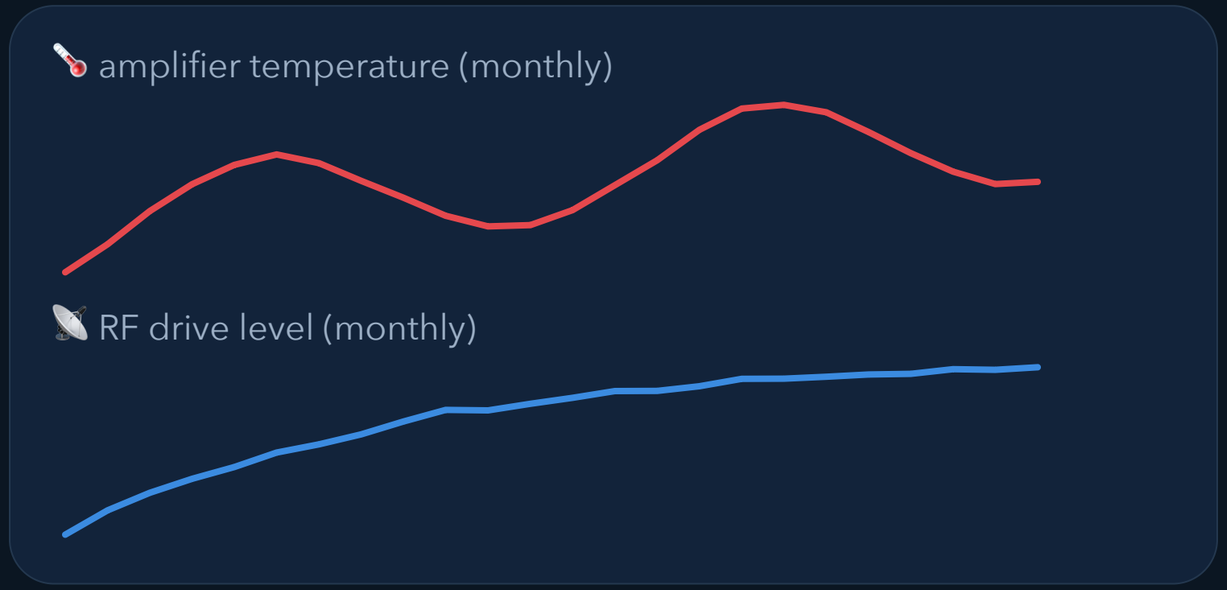
after `.diff()` (week-to-week changes)

0.06

collapses: it was just **shared trend** (Detty December 🎉)

THE REAL SATELLITE CASE · TWTA_THERMAL_DAILY.CSV

Amplifier temperature vs RF drive



levels

0.59

after `.diff()`

0.40

It **drops but survives**: part of it was shared trend, part is real physics (more drive → more heat).

REAL CORRELATION, ONE DIRECTION

Rain and uplink power: r = 0.97

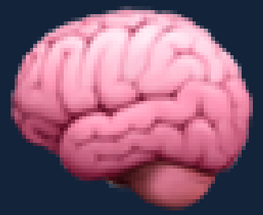


Turning up the power doesn't make it rain. 😄 Ask: **which one causes which?**

YOUR TURN · 60 MINUTES

Open the exercise notebook 👩🏾‍💻

1. Open **W2D2-exercise** in Colab
2. **File → Save a copy in Drive** 💾
3. Run **STEP 0**, then go top to bottom
4. Replace every ____, run, get the ✅



13 checks to pass

Part A: memory (lags)
Part B: the correlogram
Part C: correlation ≠ cause
🌶️ Bonus: find the yearly season

Keep the **rule book** open next to you.

TODAY IN 3 LINES

What you now know

🧠 **Autocorrelation:** how much a series matches its own copy shifted by a **lag**.

📊 **Correlogram:** spikes = season (gap = period), slow decay = trend, inside the band = noise.

🕵🏾 **Two trending series?** `.diff()` first. Correlation ≠ cause, and direction matters.

Next time 👉 **Lab Day + Quiz 1**, and you pick your final project. 🎯